

Lenovo XClarity Administrator

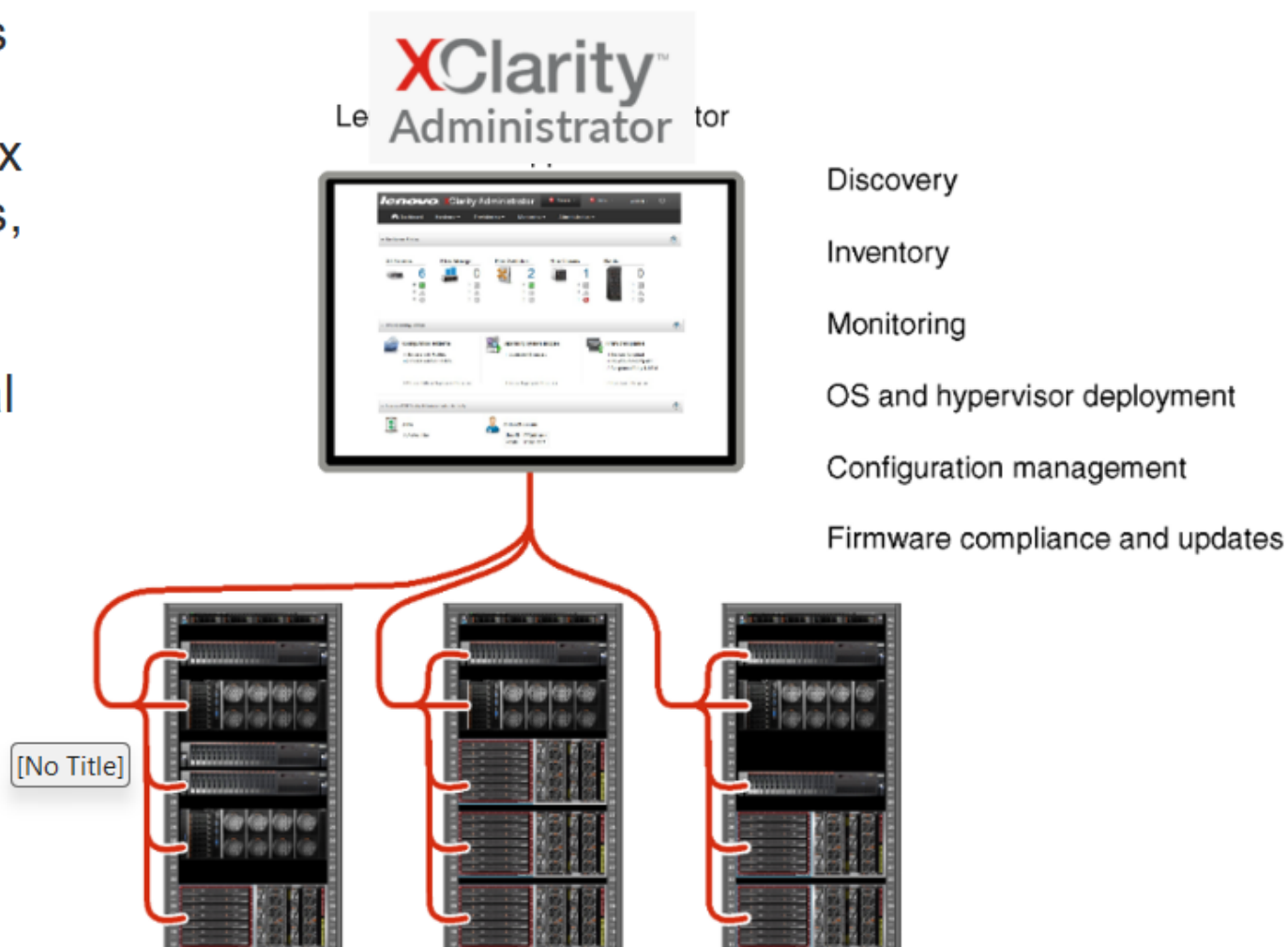
LXCA overview and how-to instructions for common tasks

Lenovo

What is Lenovo XClarity Administrator?

Lenovo XClarity Administrator (LXCA) is a comprehensive agent-less hardware management tool that supports System x servers, ThinkSystem servers, switches, and storage systems.

LXCA does not come bundled with ThinkSystem servers. It runs as a virtual appliance that automates discovery, inventory, tracking, monitoring, and provisioning for server, network, and storage hardware in a secure environment.



When to use LXCA

LXCA is recommended for the management of multiple systems from a single interface because it offers a wide range of functions in one tool. Functions, such as deploying operating systems and updating firmware on multiple systems, can be performed efficiently with LXCA. If a system in a data center encounters an error, the user can look at the LXCA dashboard to determine which system has flagged an error message. They can then review and download the logs, and then pass them on to support personnel for further troubleshooting.

LXCA also supports Windows driver updates.

LXCA edition comparison

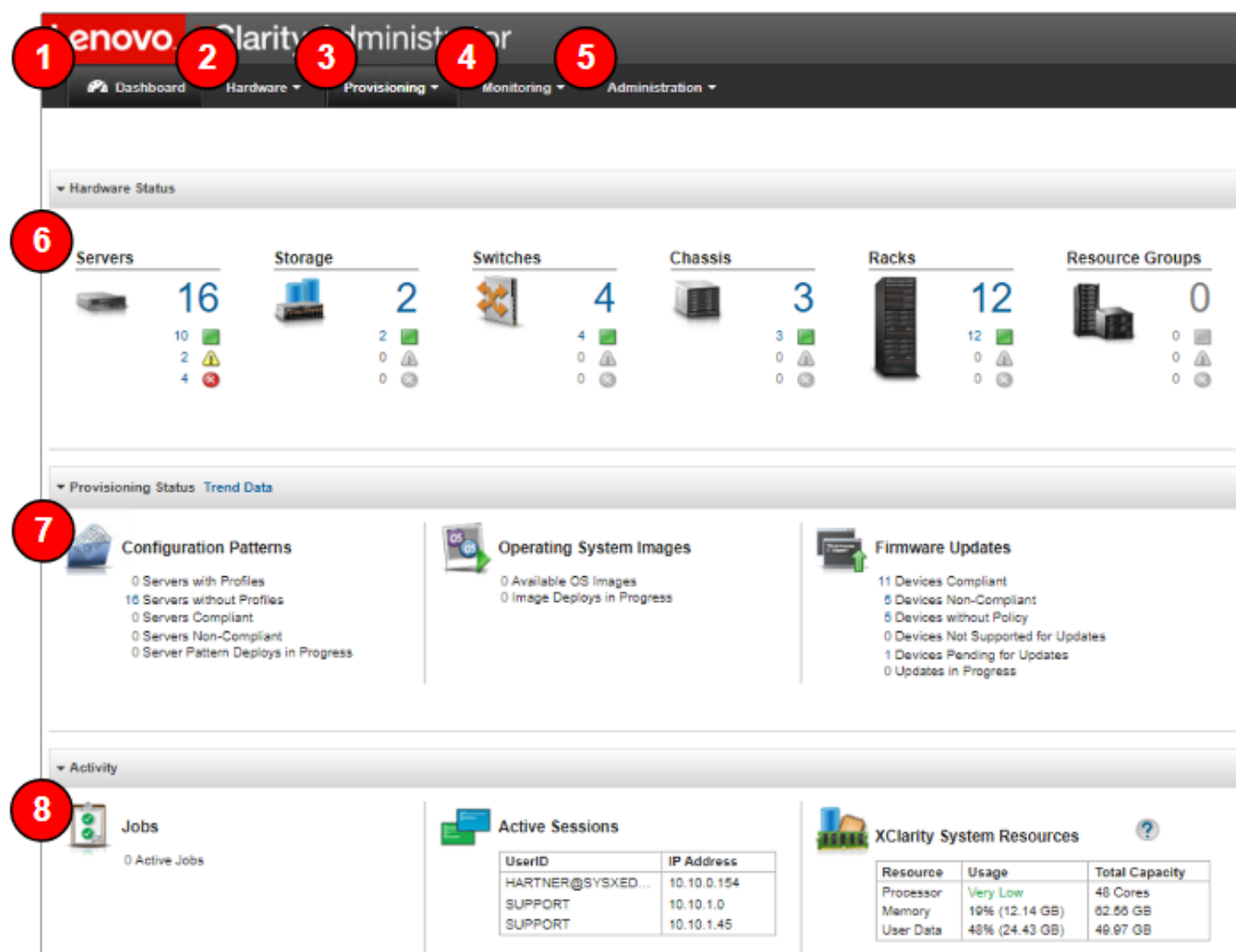
There are two LXCA editions: a free, basic edition and LXCA Pro, which requires a license upgrade.

Feature	LXCA	LXCA Pro
Licensing and support		
Free	v	
Licensed		v
Service and support		v
Key Features		
REST APIs and LXCI	v	v
Auto-discovery and asset management	v	v
Real-time monitoring, fault handling, alert notification, and Call Home	v	v
Firmware update management	v	v
Configuration patterns		v
Operating system and hypervisor installation		v

LXCA GUI quick overview

This is LXCA **Dashboard**, the LXCA homepage.

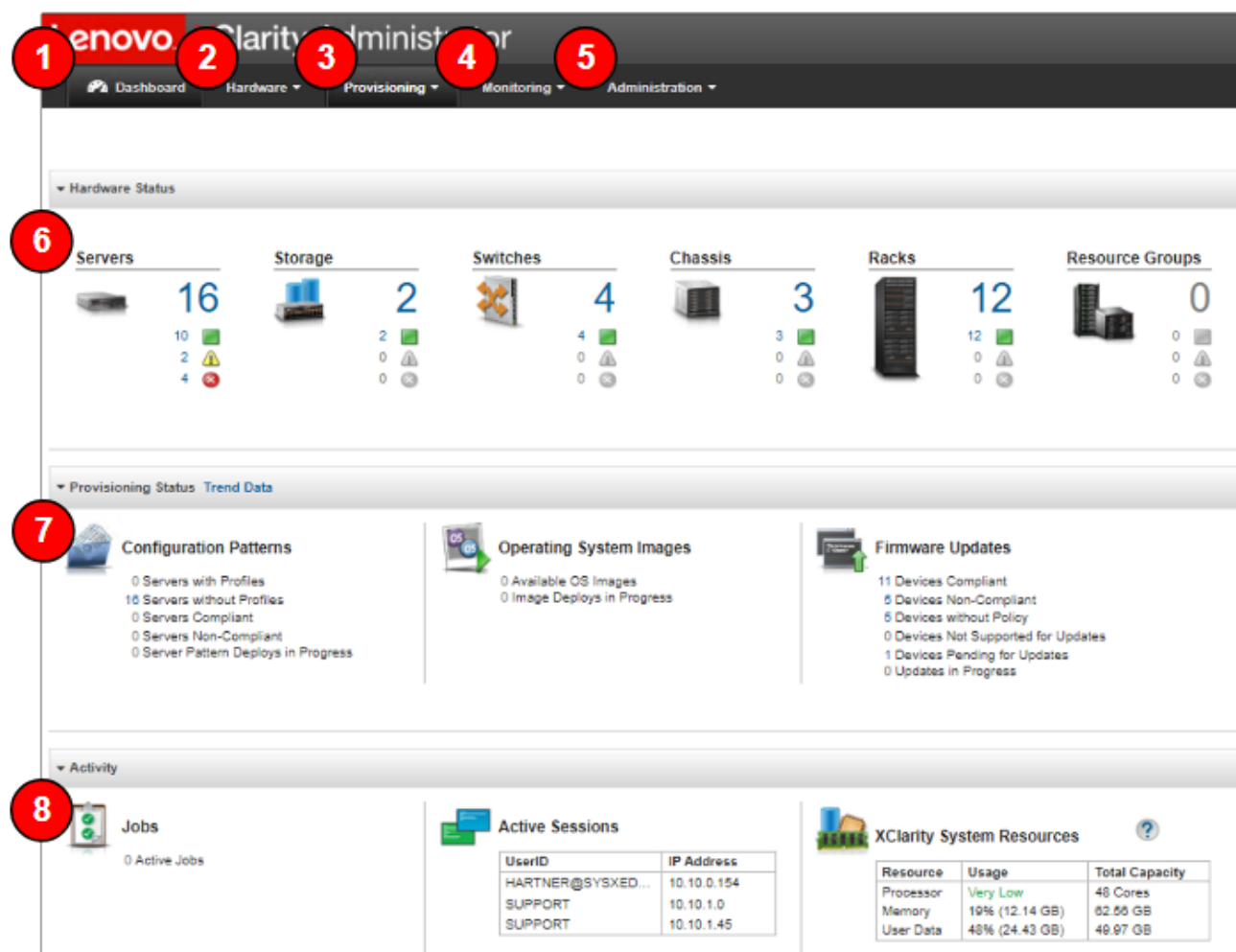
Click the icons to see descriptions of the different LXCA functions.



LXCA GUI quick overview

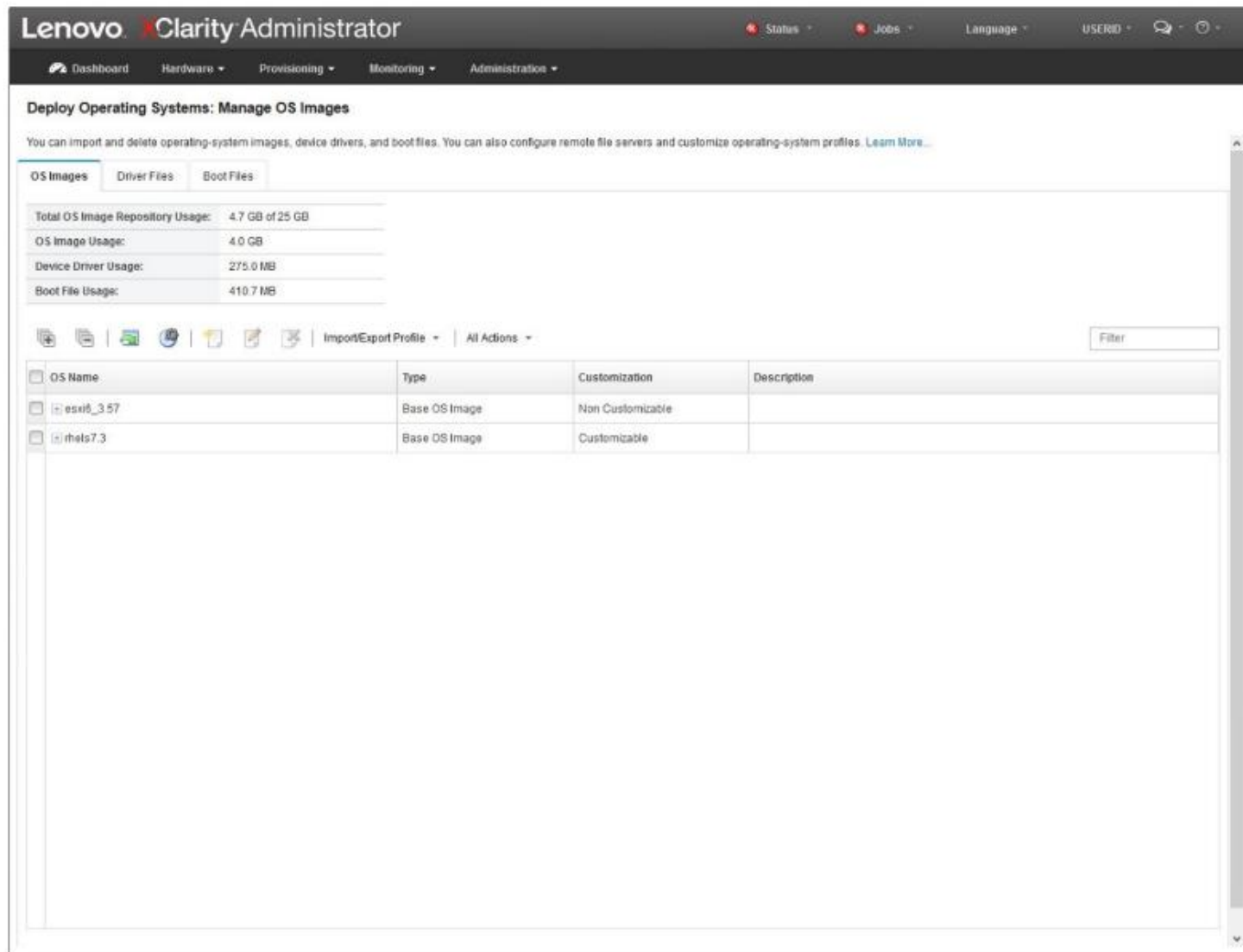
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Click the icons to see descriptions of the different LXCA functions.



OS deployment

LXCA manages a repository of operating system images that allows for the concurrent deployment of operating system images to up to 28 managed servers.



The screenshot displays the Lenovo XClarity Administrator web interface. The top navigation bar includes links for Dashboard, Hardware, Provisioning, Monitoring, and Administration. The main heading is "Deploy Operating Systems: Manage OS Images". Below this, a sub-header states: "You can import and delete operating-system images, device drivers, and boot files. You can also configure remote file servers and customize operating-system profiles. [Learn More...](#)".

There are three tabs: "OS Images" (selected), "Driver Files", and "Boot Files". Below the tabs, a summary table shows usage statistics:

Metric	Usage
Total OS Image Repository Usage:	4.7 GB of 25 GB
OS Image Usage:	4.0 GB
Device Driver Usage:	275.0 MB
Boot File Usage:	410.7 MB

Below the summary table, there is a toolbar with icons for adding, deleting, and importing/exporting profiles. A table lists the OS images:

OS Name	Type	Customization	Description
☐ esxi6_3.57	Base OS Image	Non Customizable	
☐ rhel7.3	Base OS Image	Customizable	

Firmware updates

Firmware management is simplified by assigning firmware-compliance policies to managed devices.

When users create and assign a compliance policy to managed devices, LXCA monitors changes to inventory for those devices and flags any devices that are out of compliance.

When a device is out of compliance, LXCA can be used to update it according to the policy.

Lenovo XClarity Administrator

Dashboard Hardware Provisioning Monitoring Administration

Firmware Updates: Apply / Activate

To update firmware on a device, assign a compliance policy and select Perform Updates.

Update with Policy Update without Policy

All Actions * Critical Release Information Filter By Show: All Devices Filter

Device	Rack Name / Unit	Chassis / Bay	Power	Installed Version	Assigned Compliance Policy	Compliance Target	U
3550m5-1 10.240.61.232	4 / Unit 9		On	Compliant	DEFAULT-SystemX-Servers-2017-05-02		
IMM2-402e9d8fa6 10.240.60.23	1 / Unit 9		Off	No Compliance Policy Set	No applicable policies		
sysdedudev 10.240.61.11	1 / Unit 3		On	No Compliance Policy Set	No applicable policies		
mbackup 10.240.60.112	1 / Unit 25		On	Not Compliant	DEV-Switch-SystemX-2017-05-02		
sysdeduweb 10.240.61.15	1 / Unit 5		On	No Compliance Policy Set	No applicable policies		
H05500 10.240.61.233	4 / Unit 23		On	Not Compliant	DEFAULT-SystemX-Servers-2017-05-02		
3650m5-1 10.240.61.231	4 / Unit 7		On	Not Compliant	DEFAULT-SystemX-Servers-2017-05-02		
H03500 10.240.61.234	4 / Unit 25		On	Not Compliant	DEFAULT-SystemX-Servers-2017-05-02		
H07500 10.240.61.235	4 / Unit 27		On	Not Compliant	DEFAULT-SystemX-Servers-2017-05-02		
Scalable Complex F3D836F0C5DD1...	3 / Unit 17			Not Compliant	DEV-Switch-SystemX-2017-05-02		
Scalable Complex 9B2F2A34C91311...	3 / Unit 9			Not Compliant	DEV-Switch-SystemX-2017-05-02		
Scalable Complex BB13833077D011...	3 / Unit 1			Not Compliant	DEV-Switch-SystemX-2017-05-02		
TOR-1 10.10.1.254	Unassigned / U...		On	Not Compliant	DEFAULT-CMM-switches-storage-2017-05-02	8.4.4 / 0804 Imgy_fw_torsw_g8264-8.4.4.0_a...	
RackSwitch 10.240.61.152	Unassigned / U...		On	Not Compliant	Copy-DEFAULT-CMM-switches-storage-2017-05-02	8.4.4 / 0804 Imgy_fw_torsw_g8264-8.4.4.0_a...	
S2200-2 10.240.61.156	4 / Unit 17			Compliant	DEFAULT-CMM-switches-storage-2017-05-02	GL221R019-05 / GL221R019-05 Imgy_fw_storage_gi221r019-05...	

How to download the latest firmware

All firmware files are stored and managed in the LXCA repository. Before applying any updates, the latest update files need to be downloaded into the repository.

Work through the following procedure to download the latest firmware files into the repository.

Click each number in turn to see the procedure.



Step



How to download the latest firmware

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Click each number in turn to see the procedure.

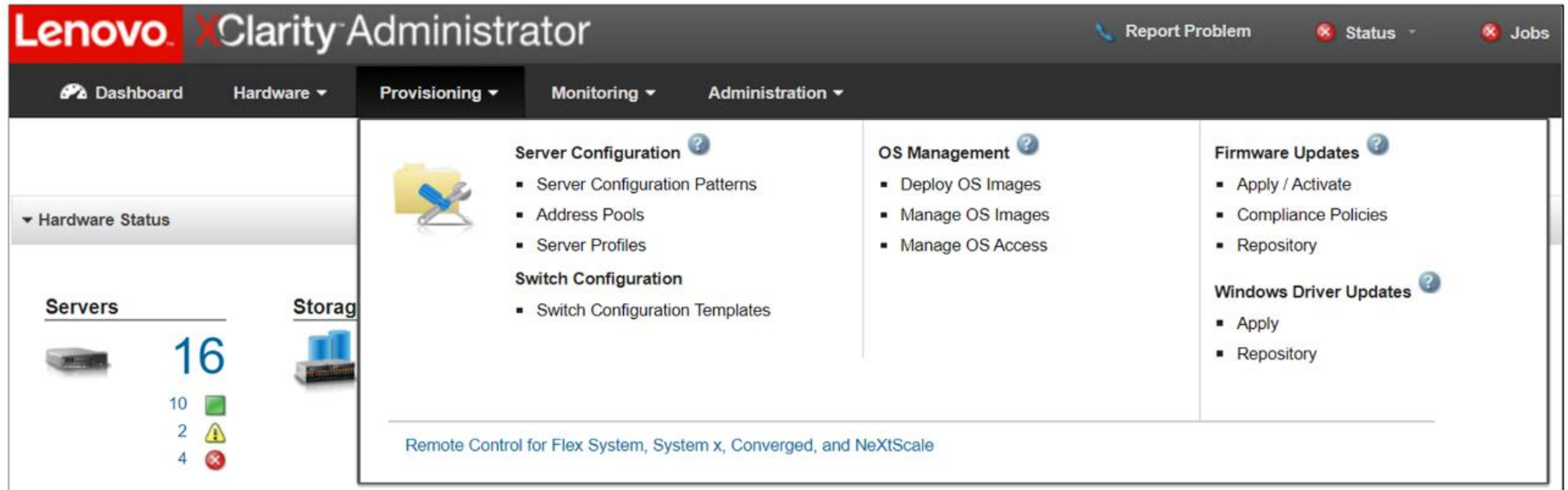


Step



How to download the latest firmware

To access the repository, click **Provisioning** in the top menu bar, and then select **Repository** from the **Firmware Updates** section.



Step 1 — 2 — 3 — 4 — 5 — 6 — 7 — 8 — 9



How to download the latest firmware

Select the system and firmware you want to download, and then click the  button.

Firmware Updates: Repository

Use Refresh Catalog to retrieve the latest product information. Ensure that you download update packages before adding the updates to a policy.

Repository Location: Local

Local repository usage: 178.03 MB of 50 GB

Individual Updates | UpdateXpress System Packs (UXSPs)

Download Selected

Product Catalog	Machine Type	Version Information	Release Date
Lenovo ThinkSystem SD630 V2 Server	7D1K		
Lenovo ThinkSystem SR550 Server	7X04		
XCC			
Lenovo XClarity Controller (XCC) Update (For AnyOS) Invgy_fw_xcc_cdi364m-5.40_anyos_noarch		5.40 / CDI364M	2020-12-01
Lenovo XClarity Controller (XCC) Update Invgy_fw_xcc_cdi334w-2.50_anyos_noarch		2.50 / CDI334W	2019-04-23
Lenovo XClarity Controller (XCC) Update Invgy_fw_xcc_cdi306x-1.08_anyos_noarch		1.08 / CDI306X	2017-11-20
UEFI			
UXPM			

Download Updates Packages

Downloading the firmware-update packages might take several minutes to complete.

Click "Download" to continue downloading the catalog, or click "Schedule" to download at a later time.

The following update packages will be downloaded.
- Invgy_fw_xcc_cdi364m-5.40_anyos_noarch.xml

Step



Individual Updates

UpdateXpress System Packs (UXSPs)

All Actions ▾ | Refresh Catalog ▾

Show: A

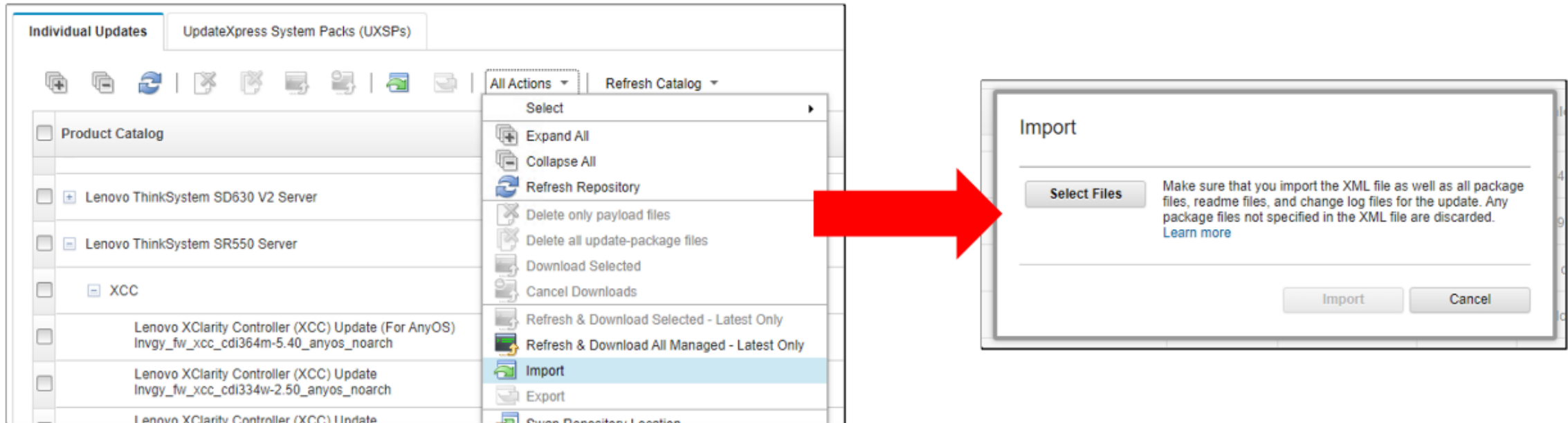
☐	Product Catalog	Machine Type	Version Information	Release Date	Download Status
☐	+ Lenovo ThinkSystem SD630 V2 Server	7D1K			44 of 44 Not Downloaded
☐	- Lenovo ThinkSystem SR550 Server	7X04			1 of 290 Downloaded
☐	- XCC				1 of 3 Downloaded
☐	Lenovo XClarity Controller (XCC) Update (For AnyOS) Invgy_fw_xcc_cdi364m-5.40_anyos_noarch		5.40 / CDI364M	2020-12-07	Downloaded
☐	Lenovo XClarity Controller (XCC) Update Invgy_fw_xcc_cdi334w-2.50_anyos_noarch		2.50 / CDI334W	2019-04-22	Not Downloaded
☐	Lenovo XClarity Controller (XCC) Update Invgy_fw_xcc_cdi306x-1.08_anyos_noarch		1.08 / CDI306X	2017-11-28	Not Downloaded



How to download the latest firmware

Users can also manually import update files into the repository by selecting **Import** from the **All Actions** drop-down menu. When importing update files, make sure that the update files include *.bin, *.xml, and *.txt files or the import will not be successful.

When the import is complete, the details of the update file will be added into the repository table.



Step



How to download the latest firmware

An alternative to downloading individual update files into the repository is to apply repository packs that include a bundle of available firmware update files and refreshed firmware-compliance policies.

Repository packs can be found on Lenovo Support. For each repository pack, all four files need to be downloaded and imported.

ⓘ

[Product Home](#)
[Drivers & Software](#)
[How To's](#)
[Warranty & Services](#)
[Repair Status](#)
[Parts](#)
[Contact Us](#)

Lenovo XClarity Repository Packs

[File Naming Guide](#)

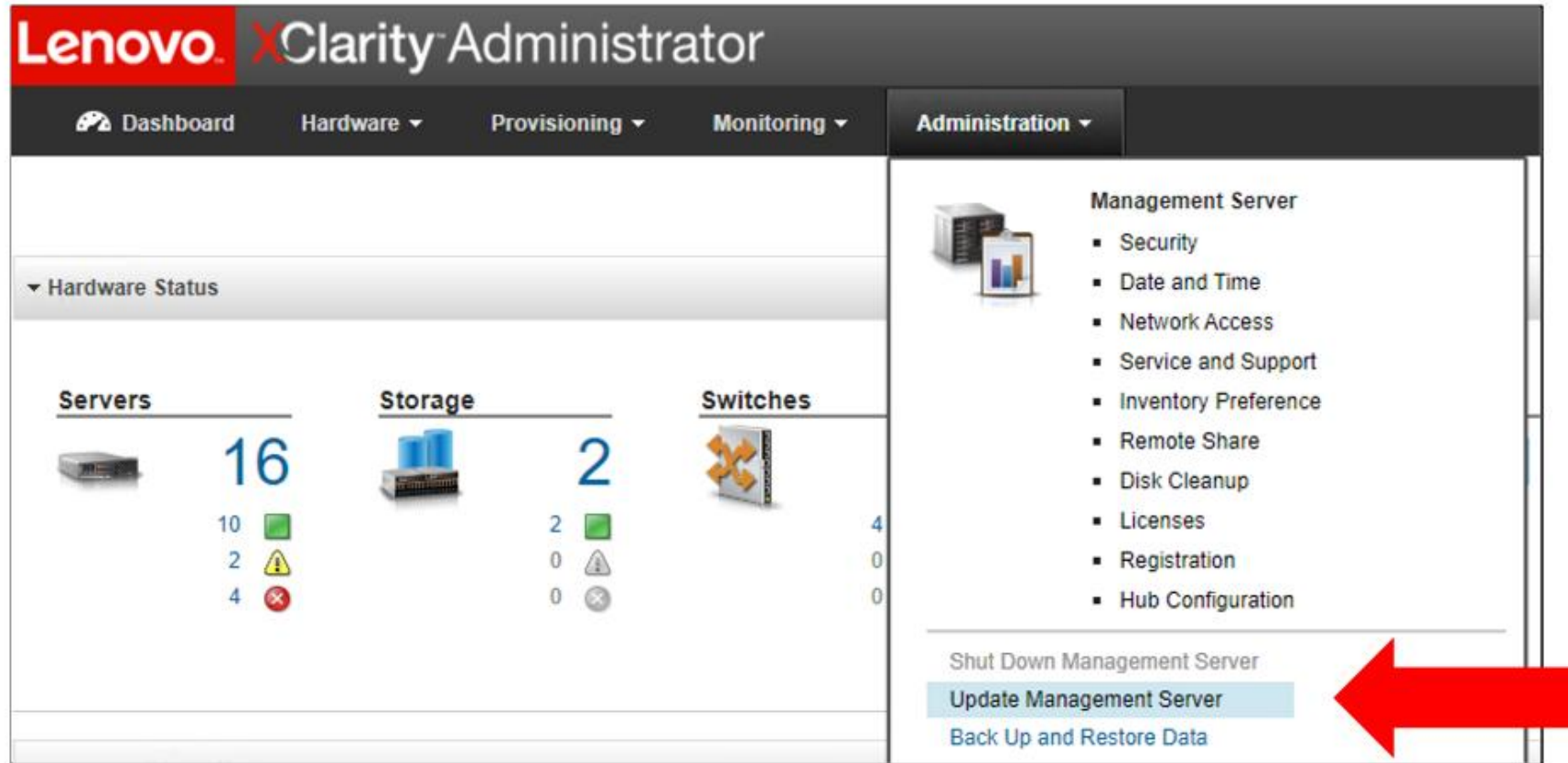
Title	Version ▼	Released ▼	Details	See Files
Lenovo XClarity Administrator cPlus ThinkServer Server Firmware Repository Pack for XClarity 3.5.0 (For AnyOS)	3.5.0 ▼	16 Feb 2022		^
Lenovo XClarity Administrator cPlus ThinkServer Server Firmware Repository Pack for XClarity 3.5.0 (For AnyOS)				
Individual Downloads		Size	Severity ⓘ	Download
Invgy_sw_lxca_cplusthinkserverrepo5-3.5.0_anyos_noarch.xml ⓘ		7.27 KB	Recommended	
Invgy_sw_lxca_cplusthinkserverrepo5-3.5.0_anyos_noarch.chg ⓘ		487 B	Recommended	
Invgy_sw_lxca_cplusthinkserverrepo5-3.5.0_anyos_noarch.txt ⓘ		1.43 KB	Recommended	
Invgy_sw_lxca_cplusthinkserverrepo5-3.5.0_anyos_noarch.tgz ⓘ		1.92 GB	Recommended	

Step



How to download the latest firmware

To import repository packs, click **Administration** in the LXCA top menu bar, and then select **Update Management Server**.

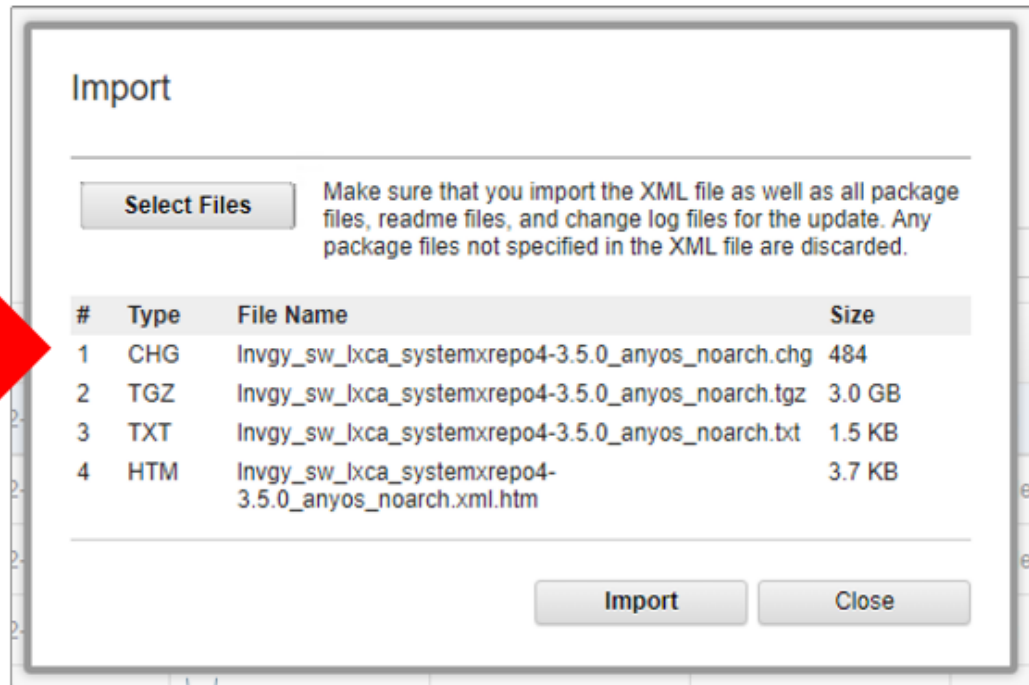
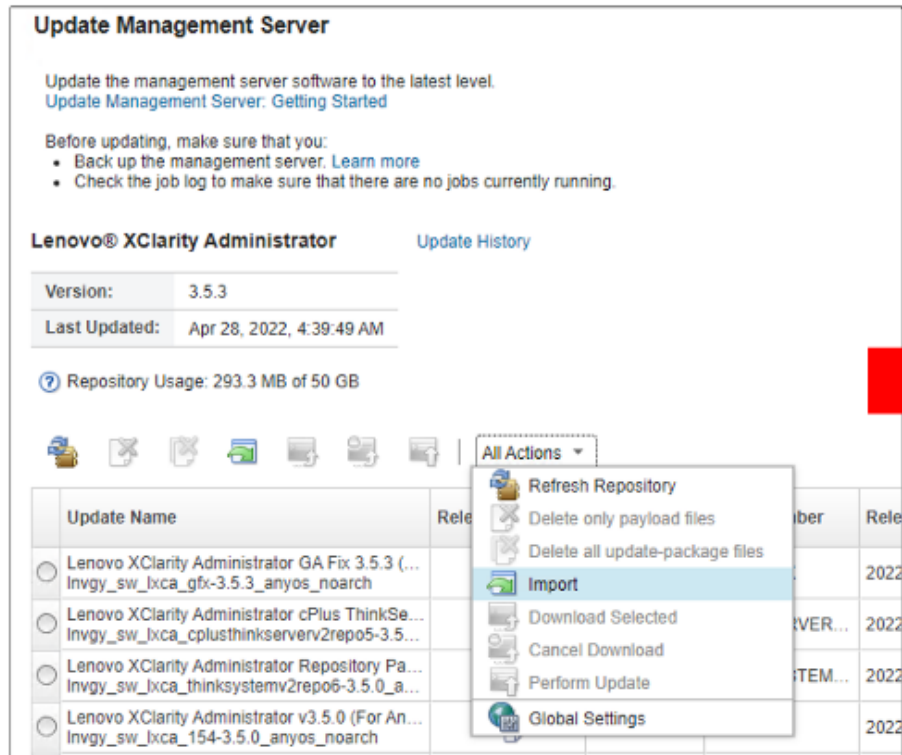


Step



How to download the latest firmware

On the **Update Management Server** page, select **Import** from the **All Actions** drop-down menu to import all four repository pack files in a single import action.

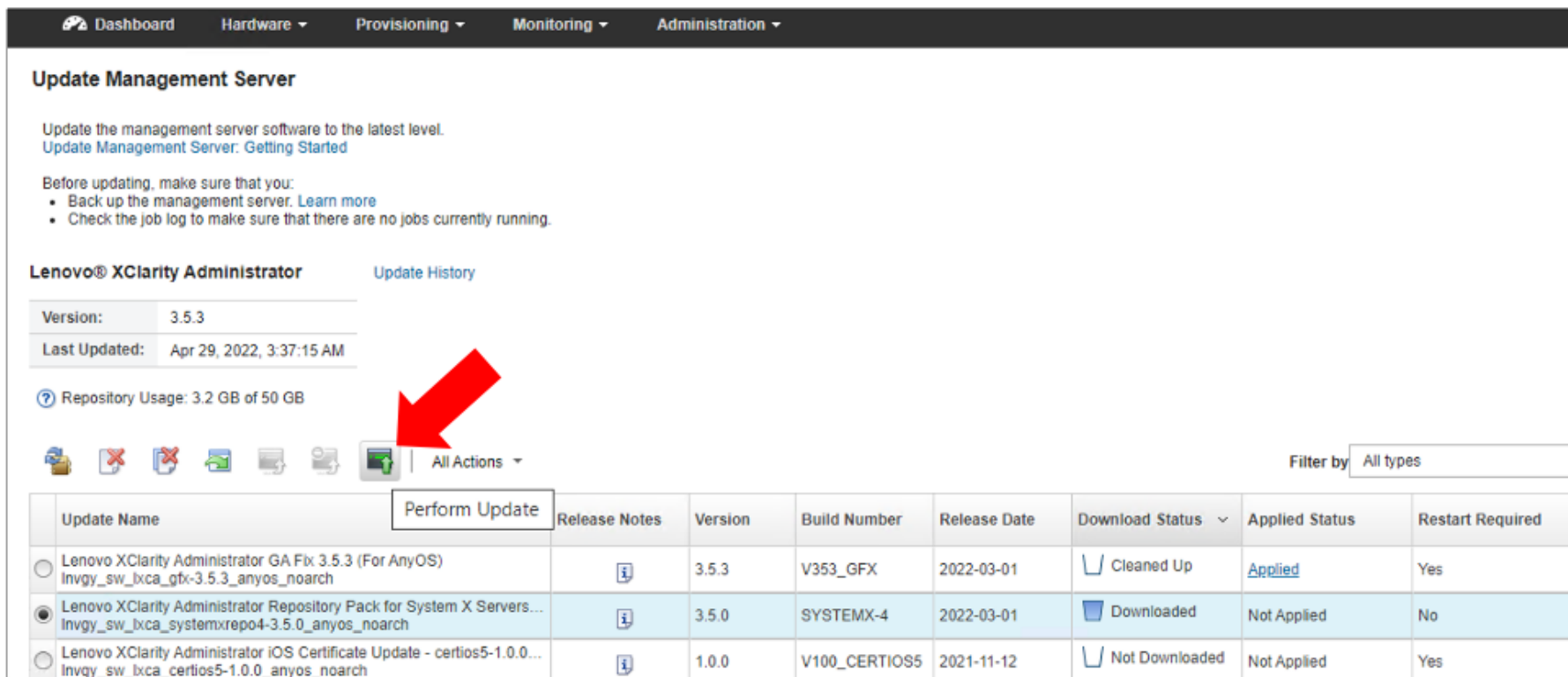


Step



How to download the latest firmware

To update the firmware, select the repository pack from the update file list, and then click the **Perform Update** button.



The screenshot shows the 'Update Management Server' section of the Lenovo XClarity Administrator interface. It includes instructions for updating the management server software and a list of available updates. A red arrow points to the 'Perform Update' button in the 'All Actions' dropdown menu.

Update Management Server

Update the management server software to the latest level.
[Update Management Server: Getting Started](#)

Before updating, make sure that you:

- Back up the management server. [Learn more](#)
- Check the job log to make sure that there are no jobs currently running.

Lenovo® XClarity Administrator [Update History](#)

Version: 3.5.3
Last Updated: Apr 29, 2022, 3:37:15 AM

Repository Usage: 3.2 GB of 50 GB

[All Actions](#) Filter by: All types

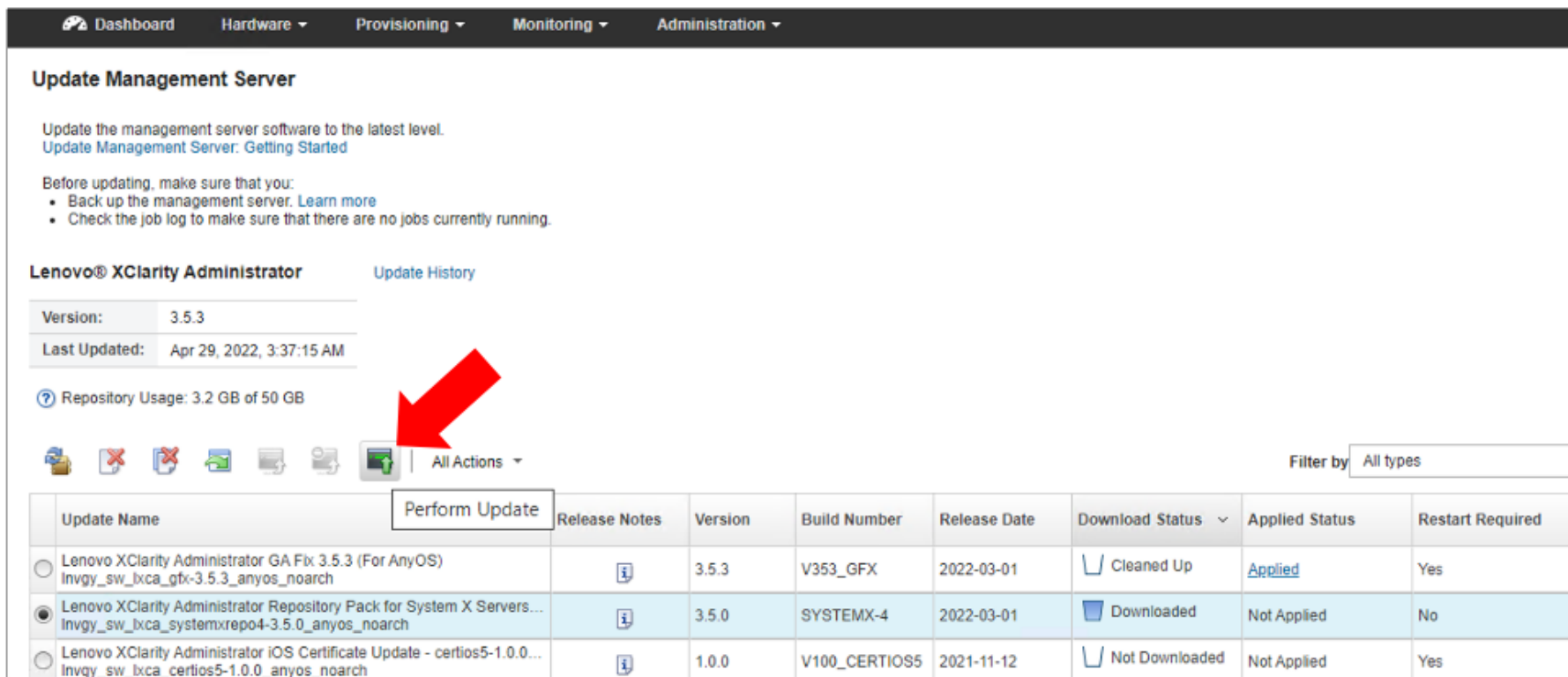
	Update Name	Release Notes	Version	Build Number	Release Date	Download Status	Applied Status	Restart Required
☐	Lenovo XClarity Administrator GA Fix 3.5.3 (For AnyOS) Invgy_sw_bxca_gfx-3.5.3_anyos_noarch	i	3.5.3	V353_GFX	2022-03-01	Cleaned Up	Applied	Yes
☒	Lenovo XClarity Administrator Repository Pack for System X Servers... Invgy_sw_bxca_systemrepo4-3.5.0_anyos_noarch	i	3.5.0	SYSTEMX-4	2022-03-01	Downloaded	Not Applied	No
☐	Lenovo XClarity Administrator iOS Certificate Update - certios5-1.0.0... Invgy_sw_bxca_certios5-1.0.0_anyos_noarch	i	1.0.0	V100_CERTIOS5	2021-11-12	Not Downloaded	Not Applied	Yes

Step



How to download the latest firmware

To update the firmware, select the repository pack from the update file list, and then click the **Perform Update** button.



The screenshot shows the 'Update Management Server' section of the Lenovo XClarity Administrator interface. It includes a navigation bar with 'Dashboard', 'Hardware', 'Provisioning', 'Monitoring', and 'Administration'. Below the navigation bar, there's a section for 'Update Management Server' with instructions on how to update the management server software. A table lists available updates, and a red arrow points to the 'Perform Update' button in the table header.

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Update the management server software to the latest level.
[Update Management Server: Getting Started](#)

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Lenovo® XClarity Administrator [Update History](#)

Version: 3.5.3
Last Updated: Apr 29, 2022, 3:37:15 AM

Repository Usage: 3.2 GB of 50 GB

Filter by: All types

	Update Name	Release Notes	Version	Build Number	Release Date	Download Status	Applied Status	Restart Required
☐	Lenovo XClarity Administrator GA Fix 3.5.3 (For AnyOS) Invgy_sw_bxca_gfx-3.5.3_anyos_noarch	i	3.5.3	V353_GFX	2022-03-01	Cleaned Up	Applied	Yes
☒	Lenovo XClarity Administrator Repository Pack for System X Servers... Invgy_sw_bxca_systemrepo4-3.5.0_anyos_noarch	i	3.5.0	SYSTEMX-4	2022-03-01	Downloaded	Not Applied	No
☐	Lenovo XClarity Administrator iOS Certificate Update - certios5-1.0.0... Invgy_sw_bxca_certios5-1.0.0_anyos_noarch	i	1.0.0	V100_CERTIOS5	2021-11-12	Not Downloaded	Not Applied	Yes

Step



How to apply firmware

After the firmware files have been downloaded into the repository, the user should work through the following procedure to apply the update.

Click each number in turn to see the procedure.

Step



How to apply firmware

On the **Firmware Updates: Apply/Activate** page, select the system or system firmware that needs to be updated, and then click the **Perform Updates** button.

Firmware Updates: Apply / Activate

To update firmware on a device, assign a compliance policy and select Perform Updates.

Update with Policy **Update without Policy**

Filter By Show: All Devices

Perform Updates

☐	Device	Groups	Rack Name / Unit	Chassis / Bay	Power	Installed Version
☐	10.10.0.66		3 / Unit 33		On	No Compliance Policy Set
☐	HX7500 10.10.3.12		4 / Unit 27		On	Pending Activation
☐	s2200-1 10.10.0.11		4 / Unit 17			No Compliance Policy Set
☒	Chassis3 fd01:1:1:1::6:50		2 / Unit 21	Chassis 3 / Bay 1	On	2.8.0 / 1AON40A

Step



How to apply firmware

In the **Update Summary** window, the user must specify an **Update Rule** and an **Activation Rule**. The table shows a summary of the items that will be updated. To start the update process, click **Perform Update**.

Update Summary
Select your Update Rule and review your updates. Then click Perform Update.

Note: The update job will run in the background and might take several minutes to complete. Updates are performed as a job. You can go to the [Jobs](#) page to view the status of the job as it progresses.

* Update Rule: Continue on error ⓘ

* Activation Rule: Immediate activation ⓘ

☐ Force update ⓘ

☒ Install prerequisite firmware ⓘ

| All Actions ▾

Device	Rack Name / Unit	Chassis / Bay	Installed Version	Downloaded Later Versions
HX3500 10.10.3.11	4 / Unit 25			

Schedule Perform Update Close

Step



How to apply firmware

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| All Actions ▾

Device	Rack Name / Unit	Chassis / Bay	Installed Version	Downloaded Later Versions
HX3500 10.10.3.11	4 / Unit 25			

Schedule Perform Update Close

Step



User management

LXCA provides a centralized authentication server to create and manage user accounts and to manage and authenticate user credentials. The authentication server is created automatically when you start the management server for the first time. The user accounts that are created for LXCA are also used to log in to managed chassis and servers.

LXCA supports three types of authentication servers:

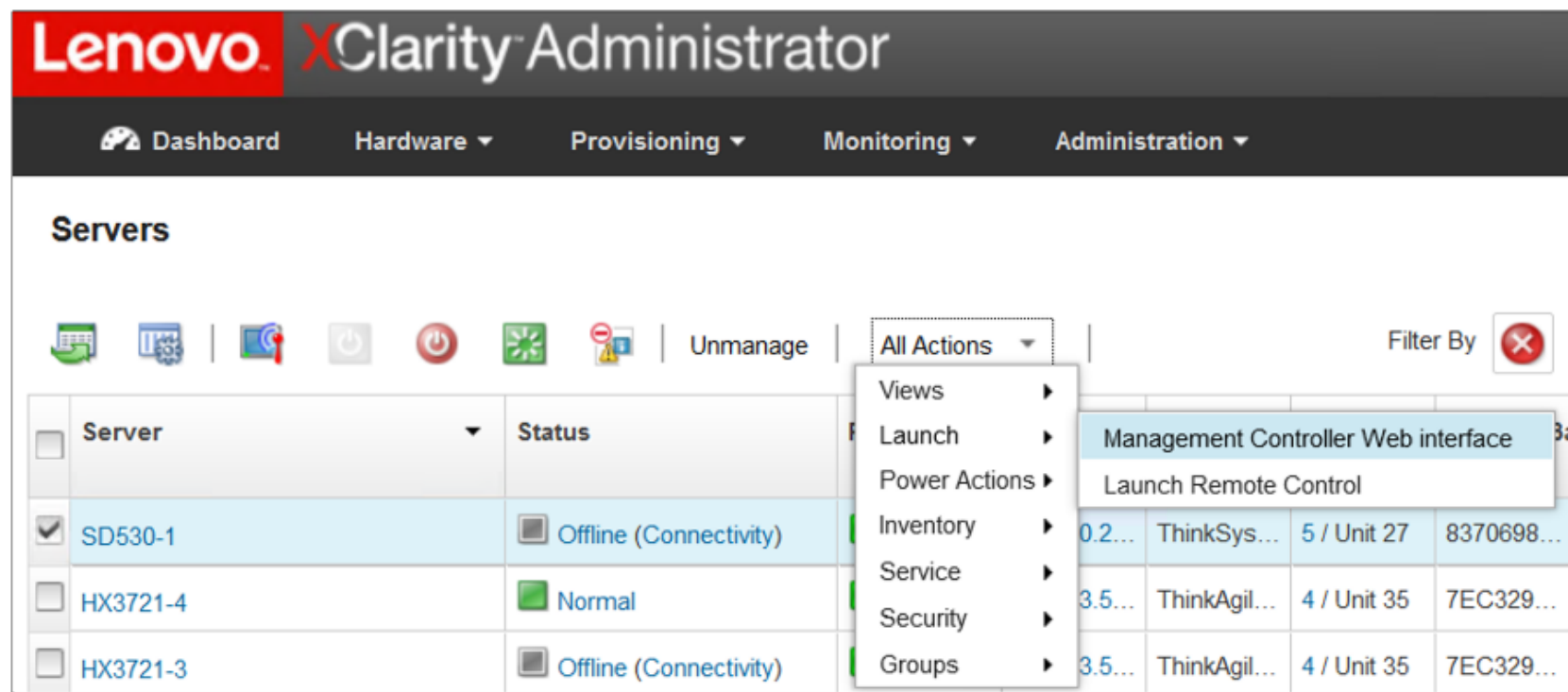
- **Local authentication server** – By default, LXCA is configured to use the local authentication server that resides on the management node.
- **External LDAP server** – At present, only Microsoft Active Directory is supported. This server must reside on an outboard Microsoft Windows server that is connected to the management network. When an external LDAP server is used, the local authentication server will be disabled.
- **External SAML 2.0 identity provider** – At present, only Microsoft Active Directory Federation Services (AD FS) is supported. In addition to entering a username and password, multi-factor authentication can be set up to enable additional security. This authentication includes a PIN code, smart card, and client certificate.

LXCA includes an audit log that provides a historical record of user actions, such as logging on, creating new users, or changing user passwords.

Click [HERE](#) to see a User Management screenshot.

XCC single sign-on through LXCA

XCC supports single sign-on through LXCA. If a server is managed by LXCA, LXCA will automatically enable the XCC single sign-on feature. When users select the server → **All Actions** → **Launch** → **Management Controller Web interface**, they will be able to directly access XCC and will not have to enter the XCC username and password.



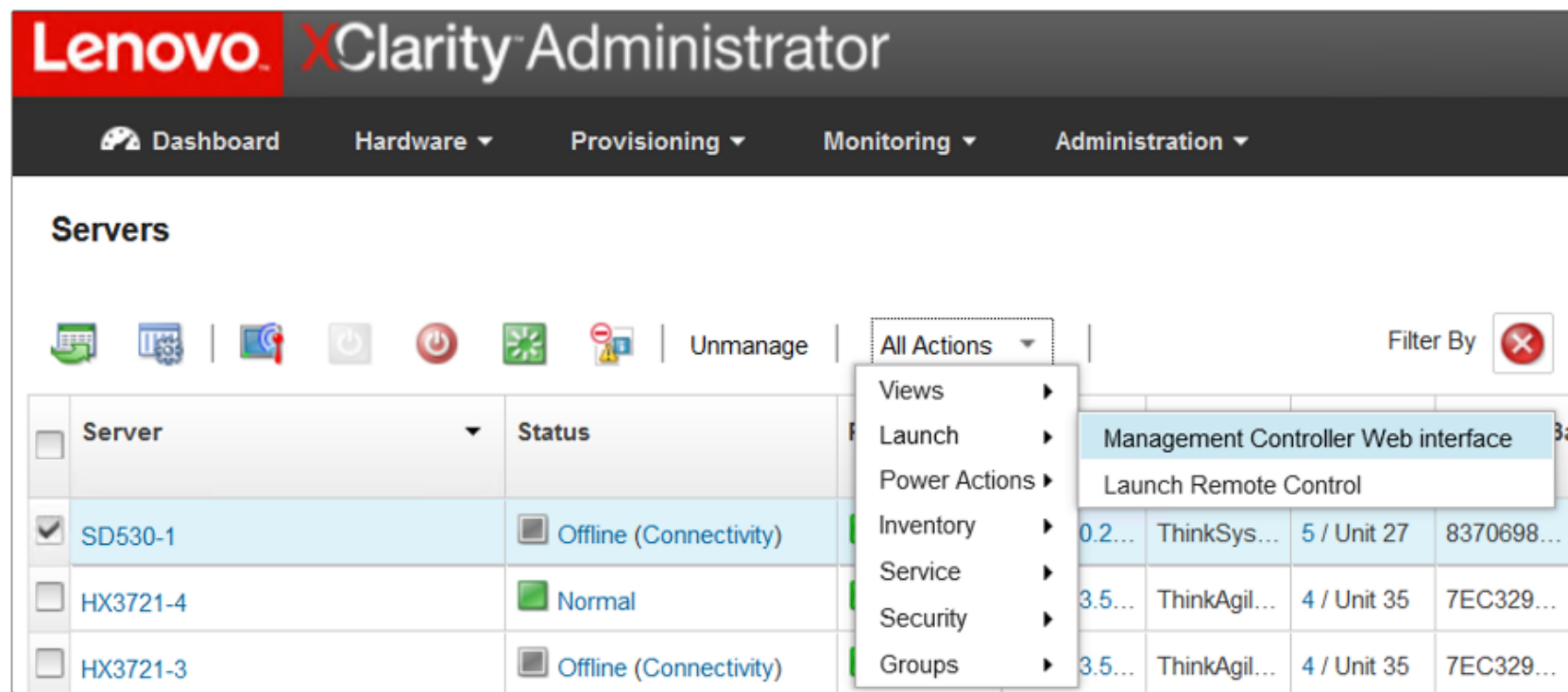
The screenshot displays the Lenovo XClarity Administrator web interface. At the top, there's a navigation bar with tabs: Dashboard, Hardware, Provisioning, Monitoring, and Administration. Below this, the 'Servers' section is active, showing a list of servers. The 'All Actions' dropdown menu is open, and the 'Launch' option is selected, which has opened a sub-menu with 'Management Controller Web interface' and 'Launch Remote Control'. The server list includes SD530-1 (Offline), HX3721-4 (Normal), and HX3721-3 (Offline).

Server	Status
SD530-1	Offline (Connectivity)
HX3721-4	Normal
HX3721-3	Offline (Connectivity)

Note: Single sign-on is supported for all ThinkSystem and ThinkAgile servers except the ThinkSystem SR635 and SR655.

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The screenshot displays the Lenovo XClarity Administrator web interface. The top navigation bar includes 'Dashboard', 'Hardware', 'Provisioning', 'Monitoring', and 'Administration'. The main section is titled 'Servers'. Below the title, there are several icons representing different server actions: a green checkmark, a blue gear, a blue key, a power button, a red power button, a green fan, a red warning icon, and an 'Unmanage' button. A dropdown menu labeled 'All Actions' is open, showing options: 'Views', 'Launch', 'Power Actions', 'Inventory', 'Service', 'Security', and 'Groups'. The 'Launch' option is selected, and a sub-menu is visible with 'Management Controller Web interface' and 'Launch Remote Control'. Below the menu, a table lists servers with columns for 'Server', 'Status', and other details. The first server, 'SD530-1', is selected and has a status of 'Offline (Connectivity)'. The second server, 'HX3721-4', has a status of 'Normal'. The third server, 'HX3721-3', has a status of 'Offline (Connectivity)'.

Server	Status
SD530-1	Offline (Connectivity)
HX3721-4	Normal
HX3721-3	Offline (Connectivity)

Note: Single sign-on is supported for all ThinkSystem and ThinkAgile servers except the ThinkSystem SR635 and SR655.

How to identify errors in LXCA

Alerts

? Alerts indicate hardware or management conditions that need investigation and user action.

      | All Actions ▾ | Excluded alerts influence health status for Show:    All Alert Sources ▾

all devices. All Dates ▾

☐	Severity ▾	Serviceability	Date and Time	Source	Alert
☐	 Critical	 User	Nov 22, 2021, 12:21:42 PM	HX3721-1	Sensor PSU2 IN Failure has transitioned to critical from a less severe state.
☐	 Critical	 User	Apr 3, 2022, 6:02:43 PM	HX3721-2	Sensor PSU1 IN Failure has transitioned to critical from a less severe state.
☐	 Critical	 User	Oct 25, 2021, 1:38:15 PM	HX3721-4	Sensor PSU2 IN Failure has transitioned to critical from a less severe state.

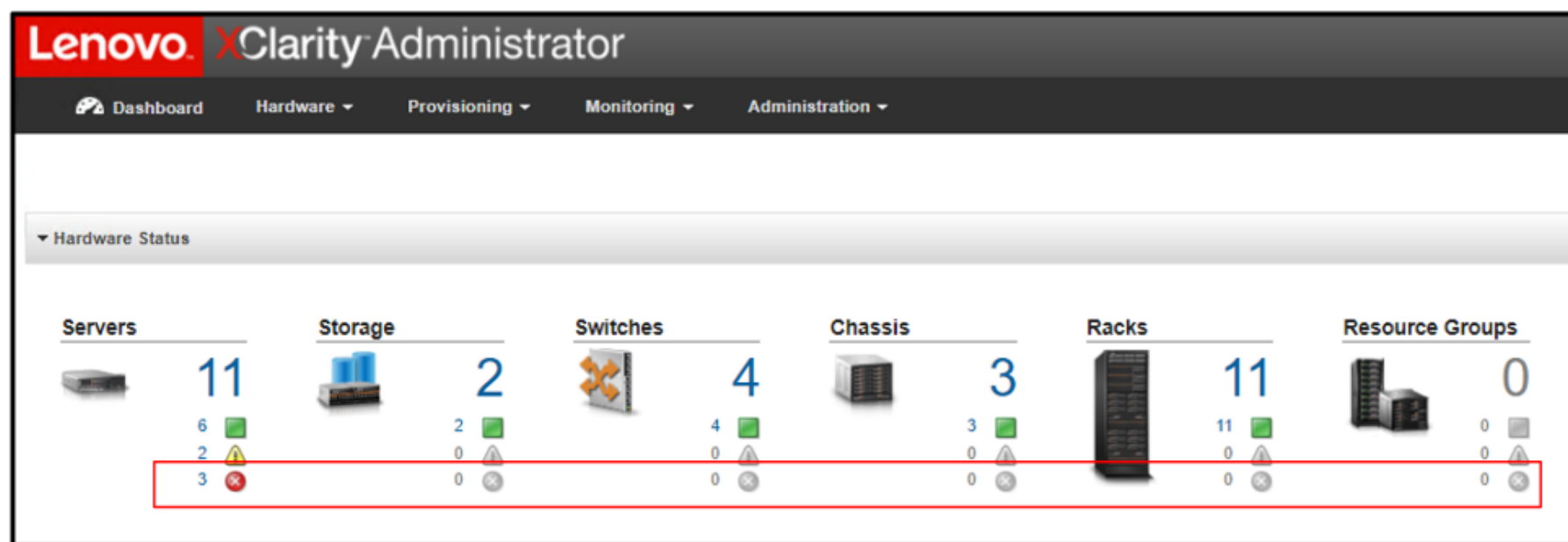
Click each number in turn to see the procedure.

Step



How to identify errors in LXCA

Error messages are accessible as soon as the user logs in to LXCA. The **Hardware Status** section of the **Dashboard** page contains a summary of the hardware monitored by LXCA. The hardware is grouped according to type, and in each group, LXCA provides a total number of event messages, units operating normally, warning messages, and critical messages. To see more detailed information about critical messages, click the number next to the  icon.



Step



How to identify errors in LXCA

After clicking the icon, a table showing the systems with critical messages will be displayed. To see more information about the critical error, click the **Critical** status.

The user will be directed to a dedicated system page with more detailed information about the critical message. Move the mouse over the error message to see more information.

The screenshot displays the Lenovo XClarity Administrator interface. The top navigation bar includes 'Dashboard', 'Hardware', 'Provisioning', 'Monitoring', and 'Administration'. The main content area is divided into two panels. The left panel, titled 'Servers', shows a table of servers with columns for 'Server', 'Status', 'Power', and 'IP Address'. The 'Status' column is highlighted with a red box, and the 'Critical' status is visible for all listed servers. The right panel, titled 'Servers>HX3721-2 Details - Alerts', shows a detailed view of a specific server's alerts. It includes a 'Severity' table with columns for 'Severity', 'Serviceability', 'Date and Time', and 'Alert'. The table shows a 'Critical' alert for 'Sensor PSU1 IN Failure has transitioned to critical from a less severe state'.

Server	Status	Power	IP Address
HX3721-2	Critical	On	10.10.3
HX3721-3	Critical	On	10.10.3
HX3721-1	Critical	On	10.10.3
HX3721-4	Critical	On	10.10.3

Severity	Serviceability	Date and Time	Alert
Critical	User	Apr 3, 2022, 6:02:43 PM	Sensor PSU1 IN Failure has transitioned to critical from a less severe state.

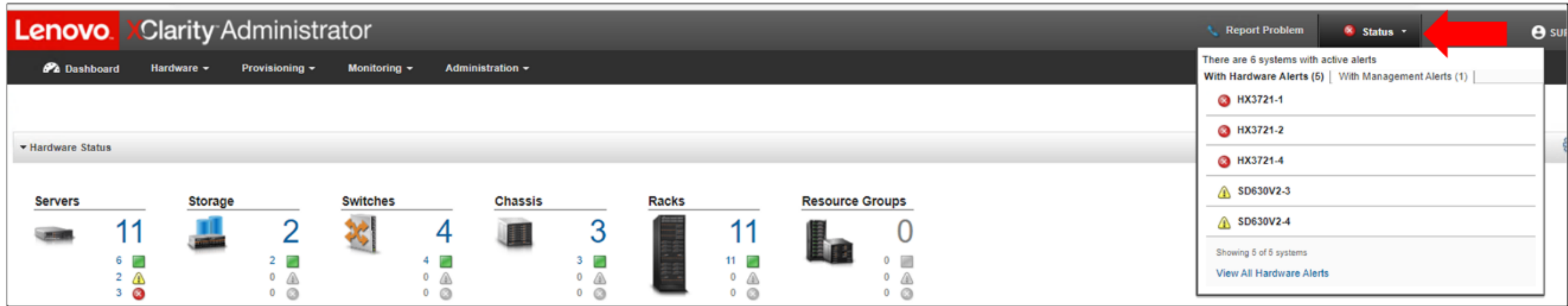
Step



How to identify errors in LXCA

Error messages can also be quickly accessed from the top menu bar on any LXCA page. Click the **Status** drop-down list to see warning or critical error messages. Move the mouse over any of the items in the list to see more information about the message.

Click **All Hardware Alerts** to go to the **Alerts** page, where all the alert messages and their descriptions are displayed. Click an alert description to see more information.



Step



How to identify errors in LXCA

Users can also click **Monitoring** in the top menu and then select **Alerts** to access alert messages. This will take the user to the same **Alerts** page shown in step 2.



Step



Service and support – Collecting service data

Select **Administration** → **Service and Support**.

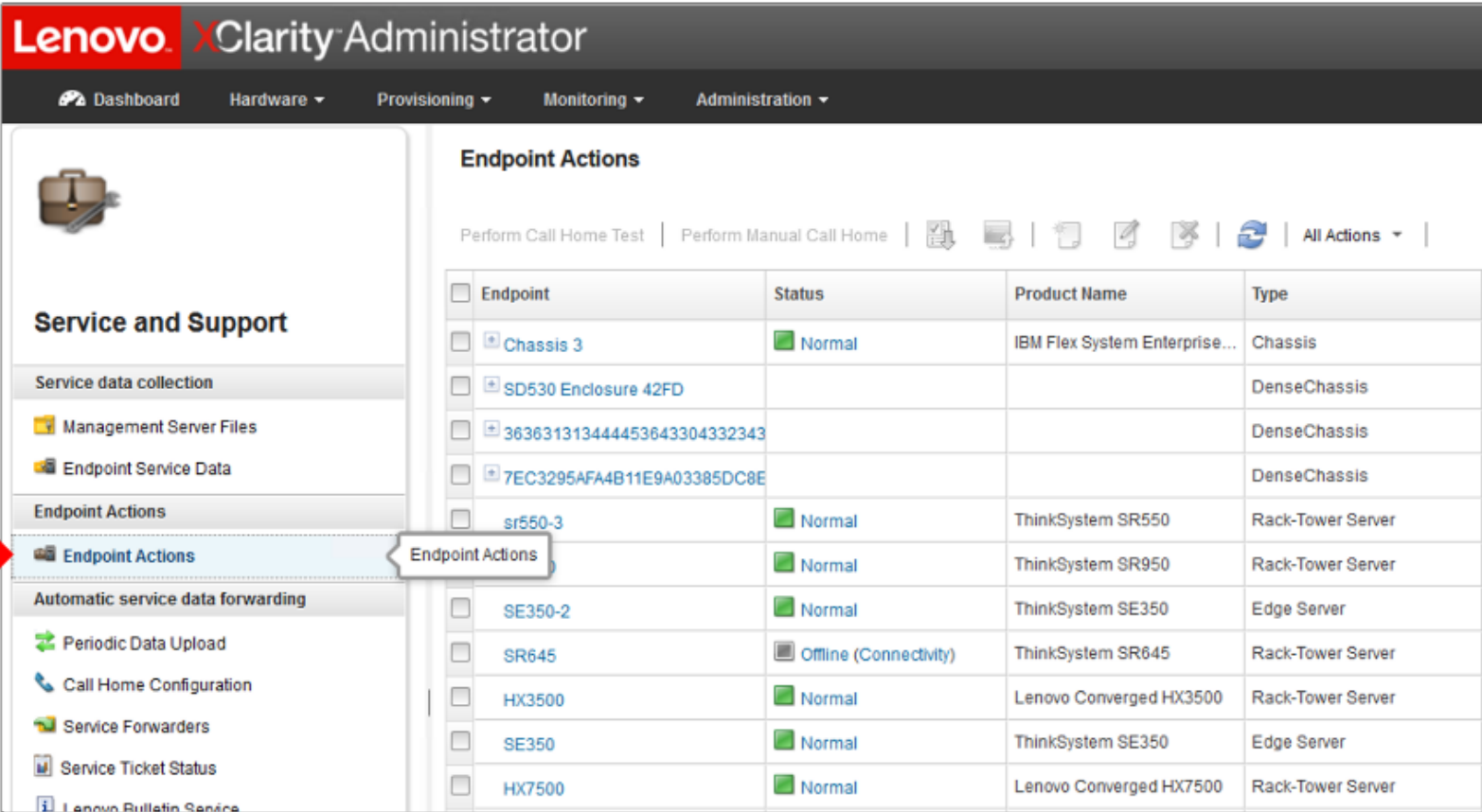


Step



Service and support – Collecting service data

Select **Endpoint Actions**.



Lenovo XClarity Administrator

Dashboard Hardware Provisioning Monitoring Administration

Service and Support

- Service data collection
 - Management Server Files
 - Endpoint Service Data
 - Endpoint Actions**
 - Automatic service data forwarding
 - Periodic Data Upload
 - Call Home Configuration
 - Service Forwarders
 - Service Ticket Status
 - Lenovo Bulletin Service

Endpoint Actions

Perform Call Home Test | Perform Manual Call Home | [Icons] | All Actions ▾

Endpoint	Status	Product Name	Type
Chassis 3	Normal	IBM Flex System Enterprise...	Chassis
SD530 Enclosure 42FD			DenseChassis
363631313444453643304332343			DenseChassis
7EC3295AFA4B11E9A03385DC8E			DenseChassis
sr550-3	Normal	ThinkSystem SR550	Rack-Tower Server
	Normal	ThinkSystem SR950	Rack-Tower Server
SE350-2	Normal	ThinkSystem SE350	Edge Server
SR645	Offline (Connectivity)	ThinkSystem SR645	Rack-Tower Server
HX3500	Normal	Lenovo Converged HX3500	Rack-Tower Server
SE350	Normal	ThinkSystem SE350	Edge Server
HX7500	Normal	Lenovo Converged HX7500	Rack-Tower Server

Step



Service and support – Collecting service data

Select one or more managed servers, and then select **Collect Service Data** from the **All Actions** drop-down menu.

The screenshot shows the Lenovo XClarity Administrator interface. The left sidebar has a 'Service and Support' section with a 'Service data collection' subsection. Under 'Endpoint Actions', the 'Endpoint Actions' link is highlighted with a red circle '1'. The main area displays a table of endpoints. The 'All Actions' dropdown menu is open, showing various actions, with 'Collect Service Data' highlighted by a red arrow and a red circle '2'.

Endpoint	Status	Product Name	Type
☐ Chassis 3	Normal	IBM Flex System Enterprise...	Chassis
☐ SD530 Enclosure 42FD			Dense
☐ 363631313444453643304332343			Dense
☐ 7EC3295AFA4B11E9A03385DC8E			Dense
☒ sr550-3	Normal	ThinkSystem SR550	Rack
☐ SR950	Normal	ThinkSystem SR950	Rack
☐ SE350-2	Normal	ThinkSystem SE350	Edge
☐ SR645	Offline (Connectivity)	ThinkSystem SR645	Rack-Tower Server

Step



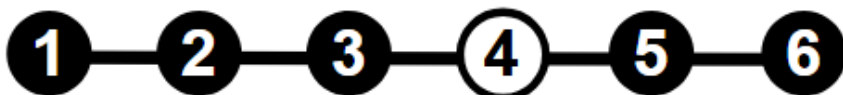
Service and support – Collecting service data

Click **Collect Service Data** to collect the data immediately.

The screenshot shows the Lenovo XClarity Administrator interface. On the left, the 'Service and Support' sidebar is expanded, showing 'Endpoint Actions' as the selected option. The main area displays a table of endpoints with columns: Endpoint, Status, Product Name, Type, and Groups. A modal dialog titled 'Collect Service Data' is open in the foreground, featuring three buttons: 'Schedule', 'Collect Service Data', and 'Cancel'. A large red arrow points from the 'Collect Service Data' button in the modal to the 'Collect Service Data' button in the table's action bar.

Endpoint	Status	Product Name	Type	Groups
Chassis 3	Normal	IBM Flex System Enterprise...	Chassis	
SD530 Enclosure 42FD			DenseChassis	
363631313444453643304332343			DenseChassis	
7EC3295AFA4B11E9A03385DC8E			DenseChassis	
sr550-3	Normal	ThinkSystem SR550	Rack-Tower Server	Casa Ce
SR950				Casa Ce
SE350-2				Casa Ce
SR645				
HX3500				
SE350	Normal	ThinkSystem SE350	Edge Server	
HX7500	Normal	Lenovo Converged HX7500	Rack-Tower Server	

Step



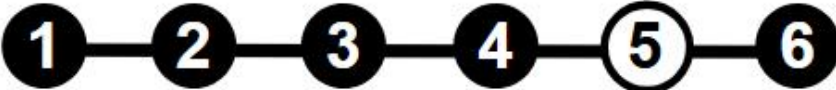
Service and support – Collecting service data

Wait for the collection process to complete.

The screenshot displays the Lenovo XClarity Administrator web interface. The left sidebar shows the 'Service and Support' section with options like 'Service data collection', 'Management Server Files', 'Endpoint Service Data', 'Endpoint Actions', and 'Automatic service data forwarding'. The main area shows the 'Endpoint Actions' table with columns for Endpoint, Status, Product Name, Type, and Groups. A modal dialog titled 'Manual collect of Service Data from device(s)' is open, showing a progress bar at 3% and a list of tasks: 'Manual collect of Service Data from device(s)' (3.00%), 'Collecting Service Data from sr550-3' (7.00%), and 'Archiving collected service data files.' (Pending). The dialog includes 'Show Details' and 'Close' buttons.

Endpoint	Status	Product Name	Type	Groups
Chassis 3	Normal	IBM Flex System Enterprise...	Chassis	
SD530 Enclosure 42FD			DenseChassis	
363831313444453643304332343			DenseChassis	
7EC3				
sr550-3				
SR95				
SE35				
SR54				
HX35				
SE35				
HX75				
HX5500	Normal	Lenovo Converged HX5500	Rack-Tower Server	

Step



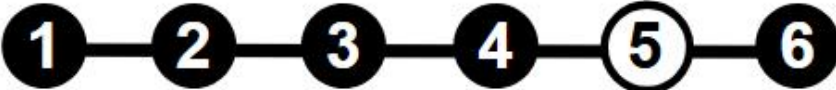
Service and support – Collecting service data

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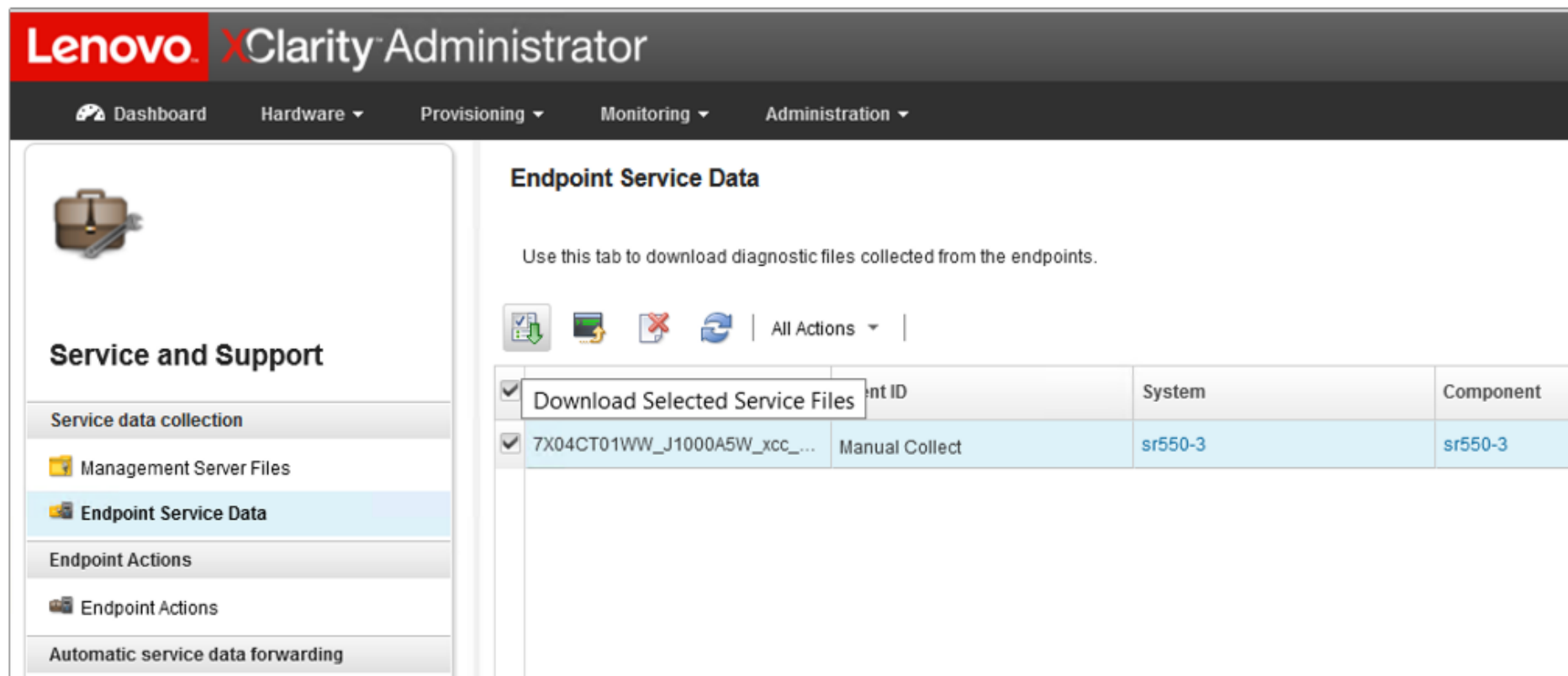
Endpoint	Status	Product Name	Type	Groups
Chassis 3	Normal	IBM Flex System Enterprise...	Chassis	
SD530 Enclosure 42FD			DenseChassis	
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SR54				
HX35				
SE35				
HX75				
HX5500	Normal	Lenovo Converged HX5500	Rack-Tower Server	

Step



Service and support – Collecting service data

After collecting the service data, go to the **Endpoint Service Data** page to download or upload the service data files.



Lenovo XClarity Administrator

Dashboard Hardware Provisioning Monitoring Administration

Service and Support

- Service data collection
 - Management Server Files
 - Endpoint Service Data**
 - Endpoint Actions
 - Endpoint Actions
 - Automatic service data forwarding

Endpoint Service Data

Use this tab to download diagnostic files collected from the endpoints.

Download Selected Service Files

Endpoint ID	System	Component
7X04CT01WW_J1000A5W_xcc_...	Manual Collect	sr550-3

Step



Task automation using scripts

LXCA can be integrated into external, higher-level management and automation platforms through open REST application programming interfaces (APIs). Using the REST APIs, LXCA can integrate with an existing management infrastructure.

The PowerShell toolkit provides a library of cmdlets to automate provisioning and resource management from a Microsoft PowerShell session. The Python toolkit provides a Python-based library of commands and APIs to automate provisioning and resource management from an OpenStack environment such as Ansible or Puppet. Both of these toolkits provide an interface to LXCA REST APIs to automate different functions such as:

- Logging in to LXCA
- Managing and unmanaging chassis, servers, storage devices, and top-of-rack switches (devices)
- Collecting and checking inventory data for devices and components
- Deploying an operating system image to one or more servers
- Configuring servers through the use of Configuration Patterns
- Applying firmware updates to devices

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- Deploying an operating system image to one or more servers
- Configuring servers through the use of Configuration Patterns
- Applying firmware updates to devices

LXCA helpful links

- LXCA quick demo:
<http://www.lenovoxclarity.com/demo/index.html>
- LXCA support resource:
<https://www.lenovo.com/tw/zh/data-center/software/systems-management/xclarity/?orgRef=https%253A%252F%252Fwww.google.com%252F#tab-ww-xclarity-main-tab-5>
- LXCA documentation:
https://sysmgt.lenovofiles.com/help/topic/com.lenovo.lxca.doc/aug_product_page.html
- LXCA video tours:
<https://datacentersupport.lenovo.com/lu/en/solutions/ht513652>